



Technische  
Universität  
Dresden



Deep Learning II

# Digitization and Data Analytics

# Structure of the Lecture Series

Lecture 1: Introduction

Lecture 2: Data Preprocessing

Lecture 3: Research Data Management

Lecture 4: Machine Learning I - Unsupervised Learning (as video)

Lecture 5: Machine Learning II - Supervised Learning (as video)

Lecture 6: Machine Learning III

Lecture 7: Data Analytics and Statistics

Lecture 8: Deep Learning I (as video)

Lecture 9: Deep Learning II

Lecture 10: Distributed Computing I

Lecture 11: Distributed Computing II

Lecture 12: Consequences

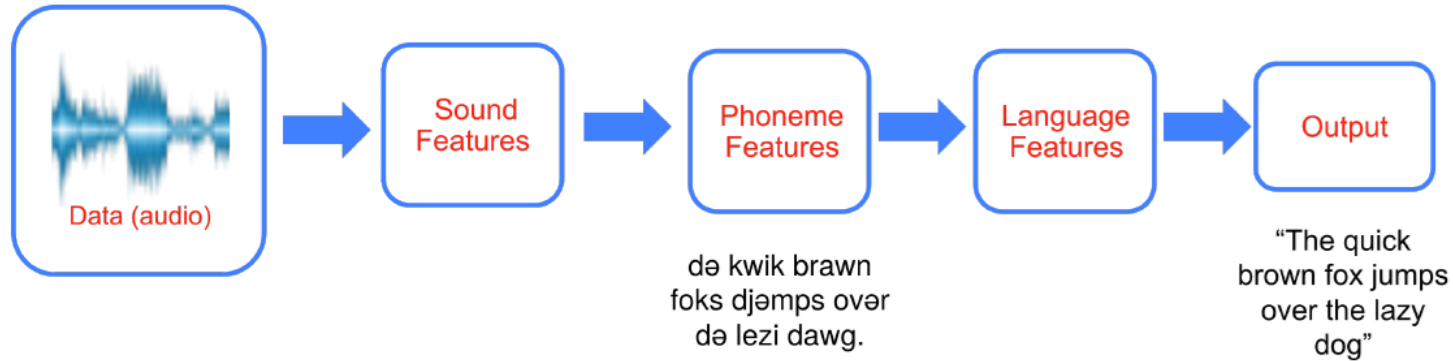
# Structure of this Lecture

## Network Architectures and Applications

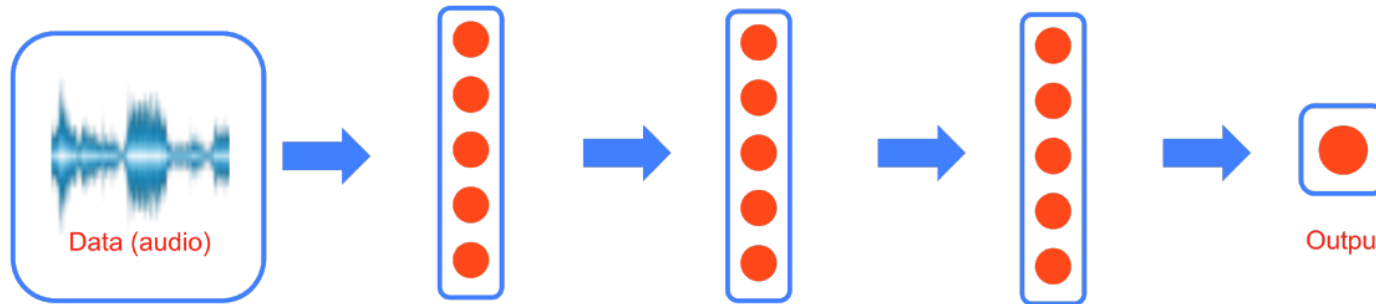
- The power of DNNs
- Images and Convolutional Neural Networks
- Sequences and Recurrent Neural Networks
- Text data and Transformers

# Speech Recognition

Speech recognition used to employ hand-crafted layers of abstraction

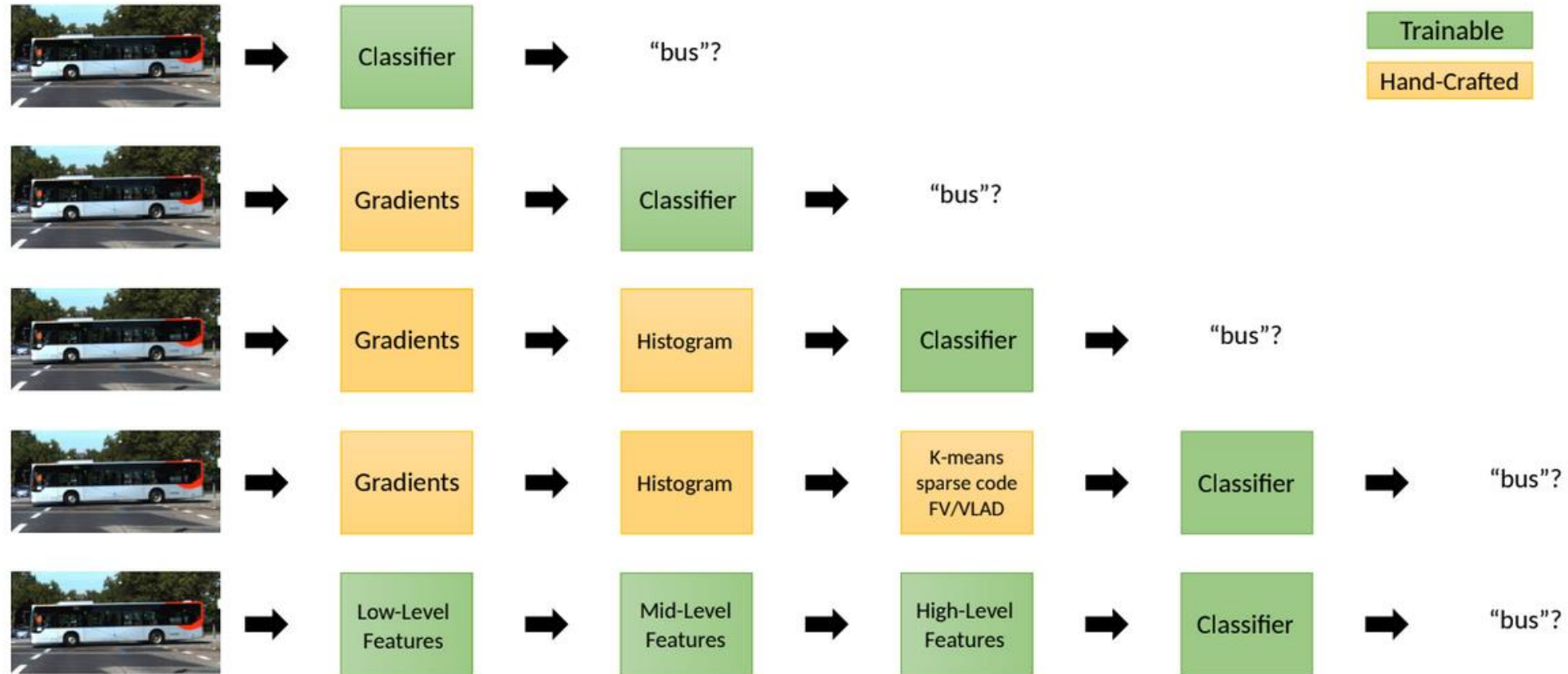


DNN learns these layers automatically, given enough data



Figures: Andrew NG @ GTC2015, Jeffrey Kelling HZDR

# Image Classification



# Public Datasets for Image Classification

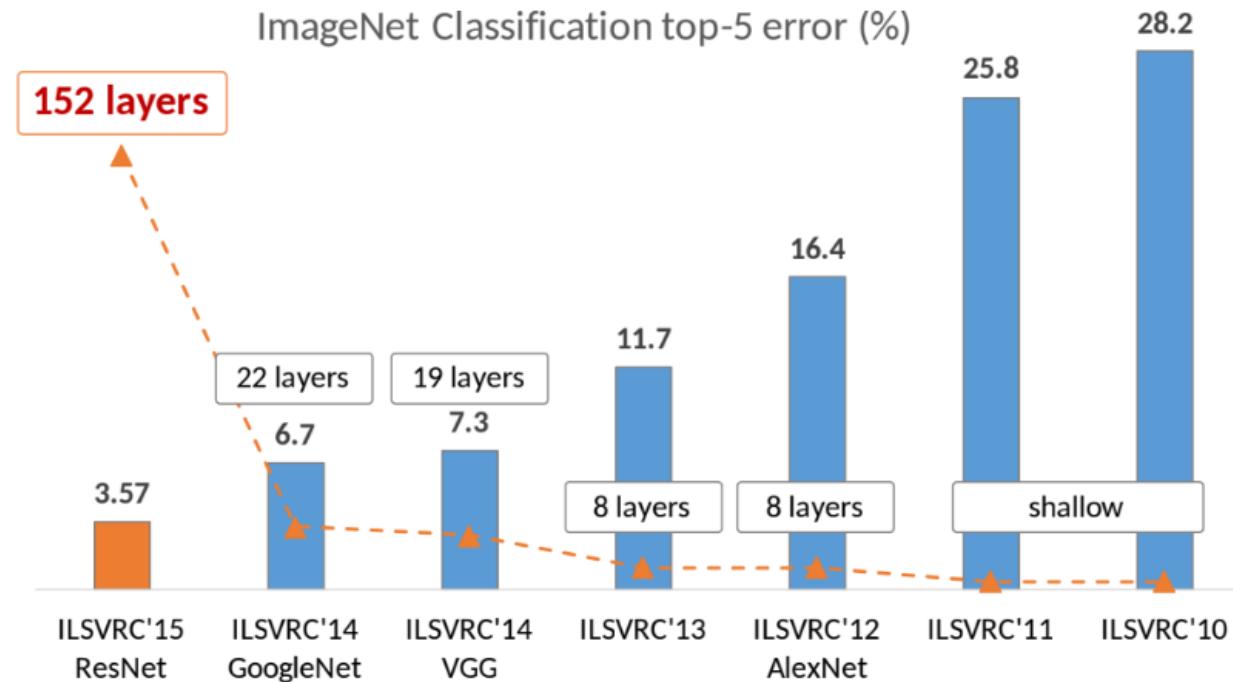
|          | MNIST                                                                             | CIFAR-10                                                                           | CIFAR-100                                                                           | ImageNet                                                                            |
|----------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|          |  |  |  |  |
| #classes | 10                                                                                | 10                                                                                 | 100                                                                                 | 1000                                                                                |
| year     | 1998                                                                              | 2009                                                                               |                                                                                     | 2010                                                                                |

ImageNet challenge [1,2] also contains object localization and scene classification

[1] <http://image-net.org>

[2] [http://rodrigob.github.io/are\\_we\\_there\\_yet/build/classification\\_datasets\\_results.html](http://rodrigob.github.io/are_we_there_yet/build/classification_datasets_results.html)

# ImageNet Large Scale Visual Recognition Challenge



Classification into 1000 object categories

Russakovsky et al.: ImageNet Large Scale Visual Recognition Challenge. IJCV, 2015.

<https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/>

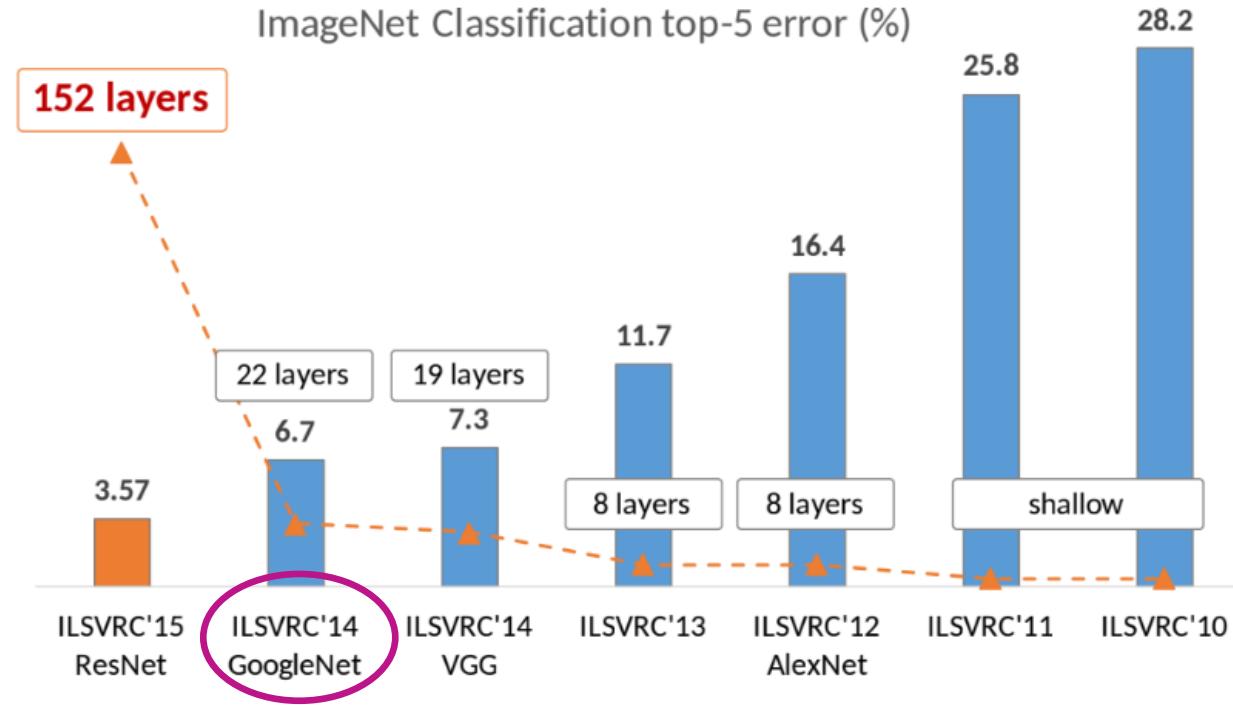
Lecture 07 Convolutional Neural Networks

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# ImageNet Large Scale Visual Recognition Challenge



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Lecture 07 Convolutional Neural Networks

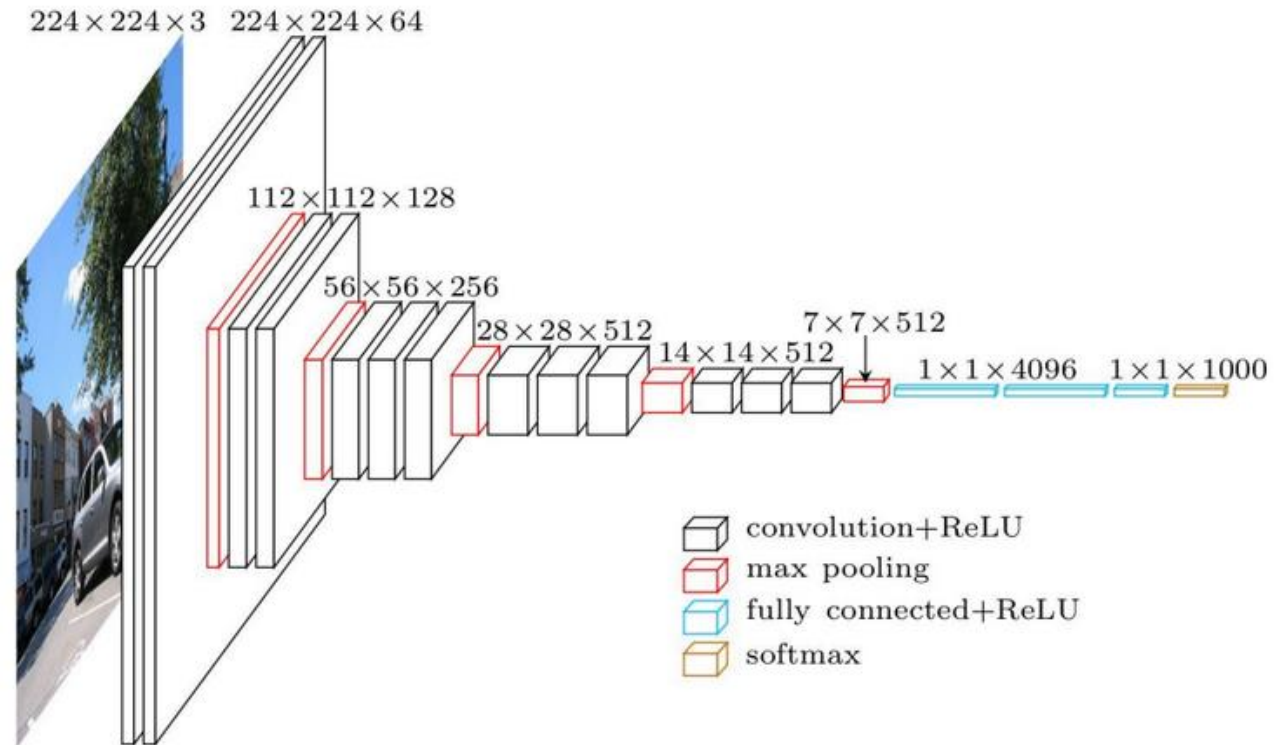
# VGG\* Network

## Types of layers

Convolution

Pooling

Fully connected



\* (Visual Geometry Group Oxford University)

[1] <https://www.kaggle.com/code/blurredmachine/vggnet-16-architecture-a-complete-guide/notebook>

# Convolutional Networks - Notation

$B$ : **Batch size**

$W$ : **Width** of image or feature map

$H$ : **Height** of image or feature map

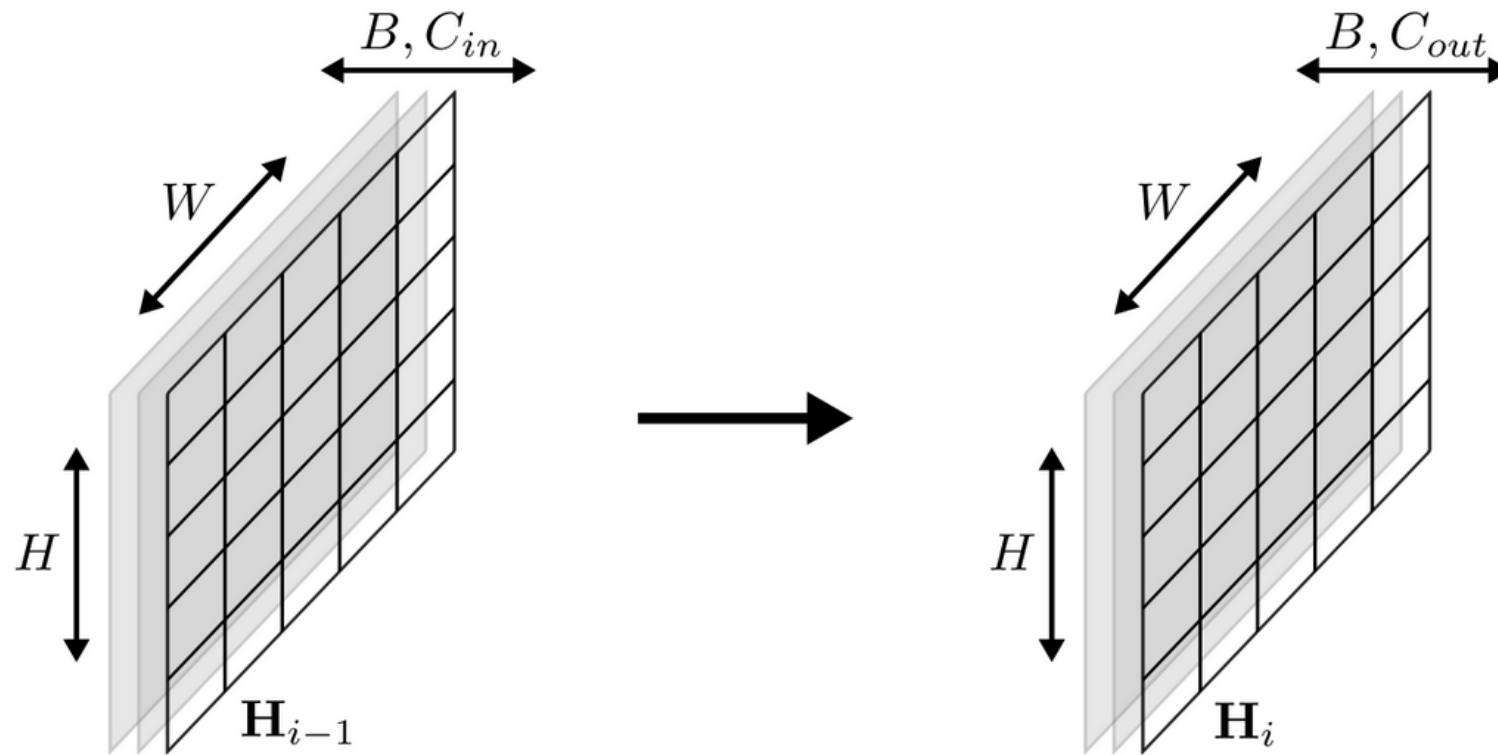
$C$ : Number of feature **channels** (= neurons per pixel)

- $C_{in}$ : Number of feature channels in current layer
- $C_{out}$ : Number of feature channels in next layer
- $C_{in} = 1$  for first layer if input is a grayscale image

$C_{in} = 3$  for first layer if input is a color image

$K \times K$ : Convolutional filter **kernel size**

# Convolutional Layer – Spatial Dimension

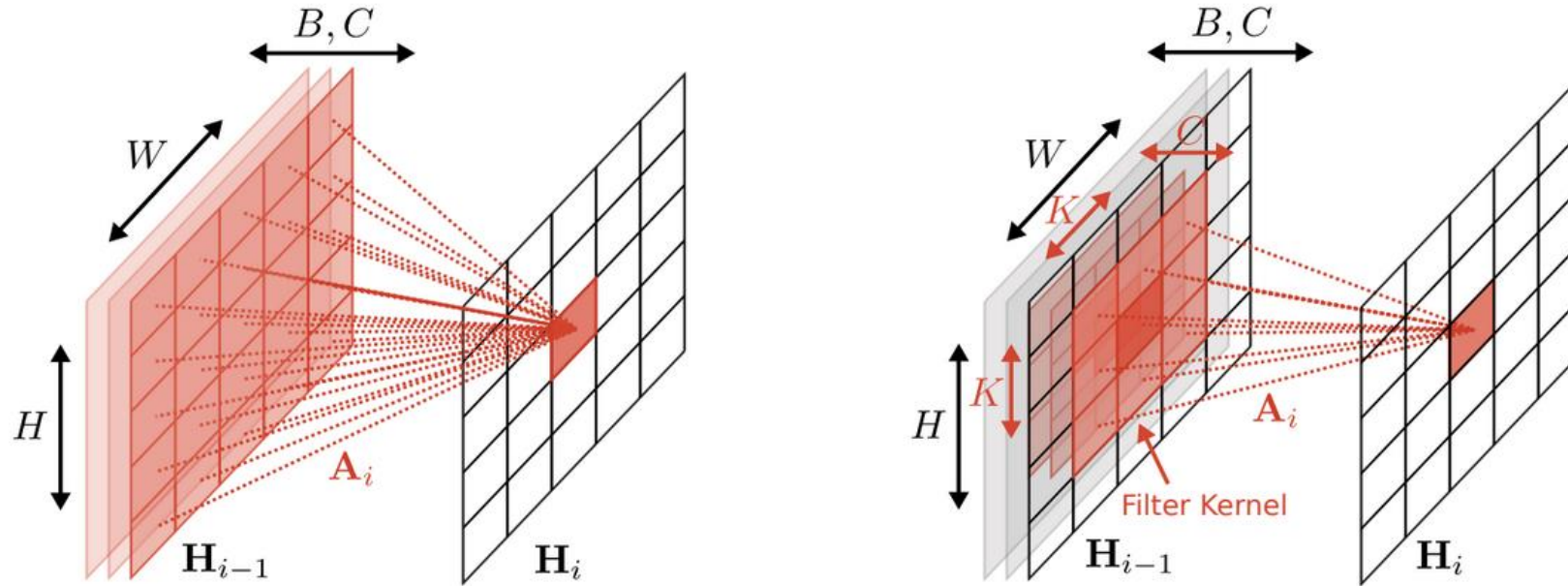


Note: In contrast to MLPs, Convolutional layers have a **spatial extent**.

<https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/>

Lecture 07 Convolutional Neural Networks

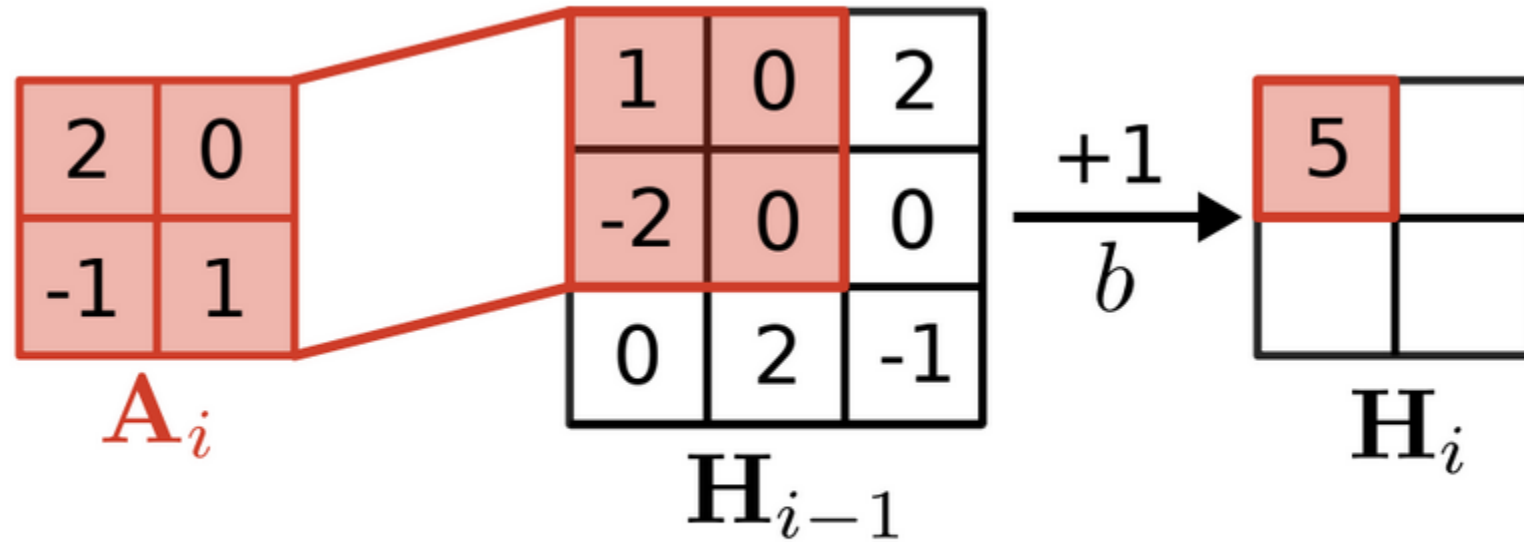
# Convolutional Layer – Weights



- **Fully connected layer:** # Weights =  $W \times H \times C_{out} \times (W \times H \times C_{in} + 1)$
- **Convolutional layer:** # Weights =  $C_{out} \times (K \times K \times C_{in} + 1)$  („weight sharing“)
- $C_{in}, C_{out}$ : input and output channels,  $W \times H$ : layer size,  $K \times K$ : kernel size
- Convolutional kernel is three-dimensional. For clarity, in the figure only one output channel is depicted.

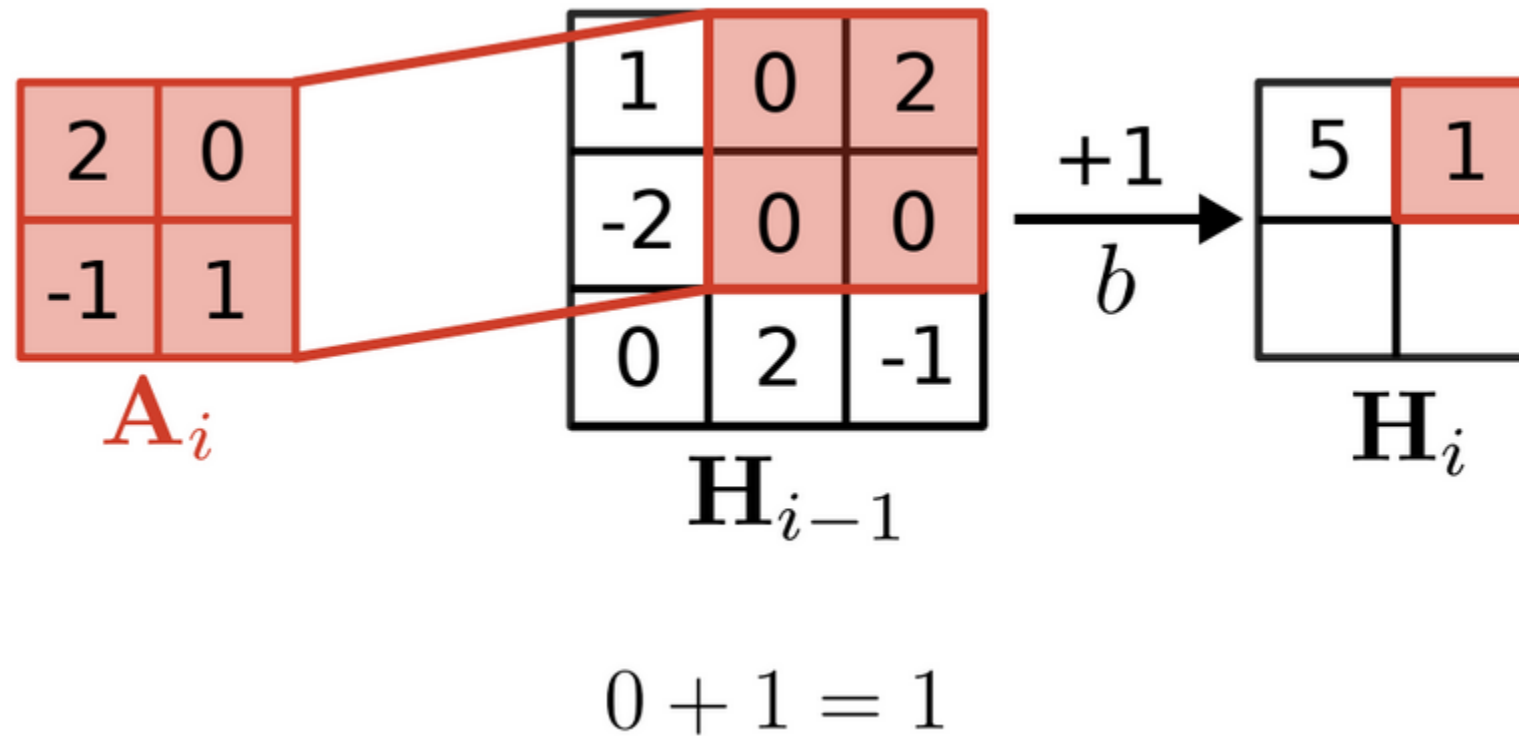
<https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/>

# Convolution Example

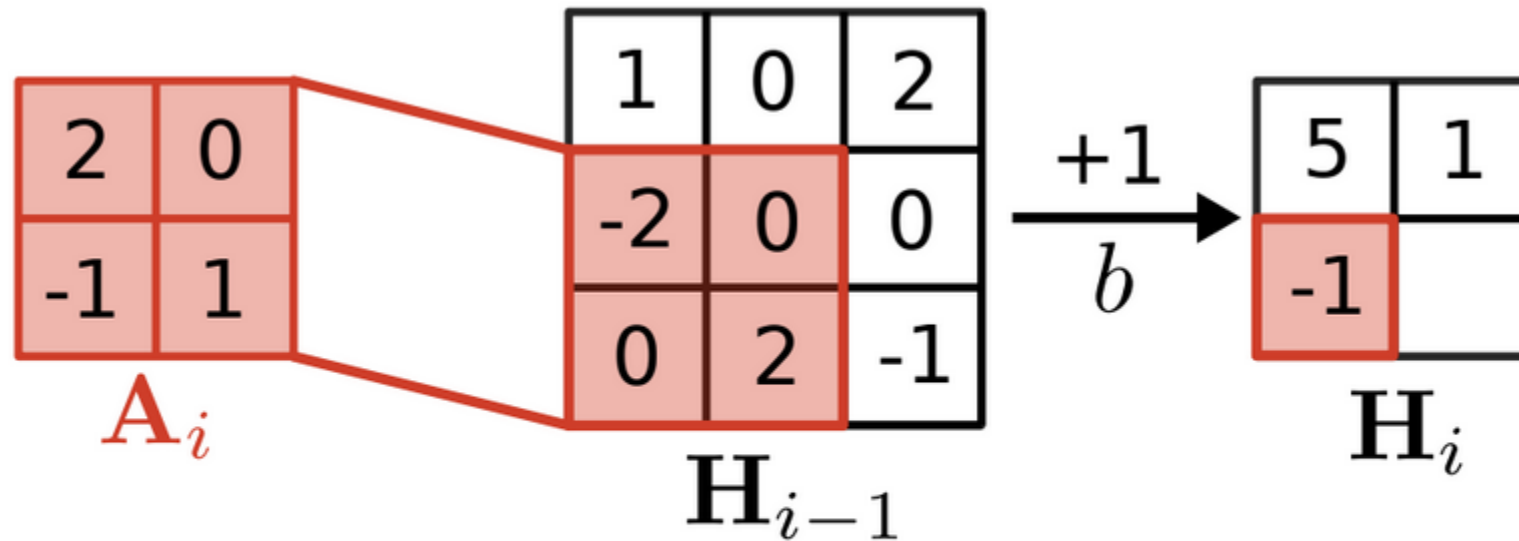


$$2 \cdot 1 + (-1) \cdot (-2) + 1 = 5$$

# Convolution Example

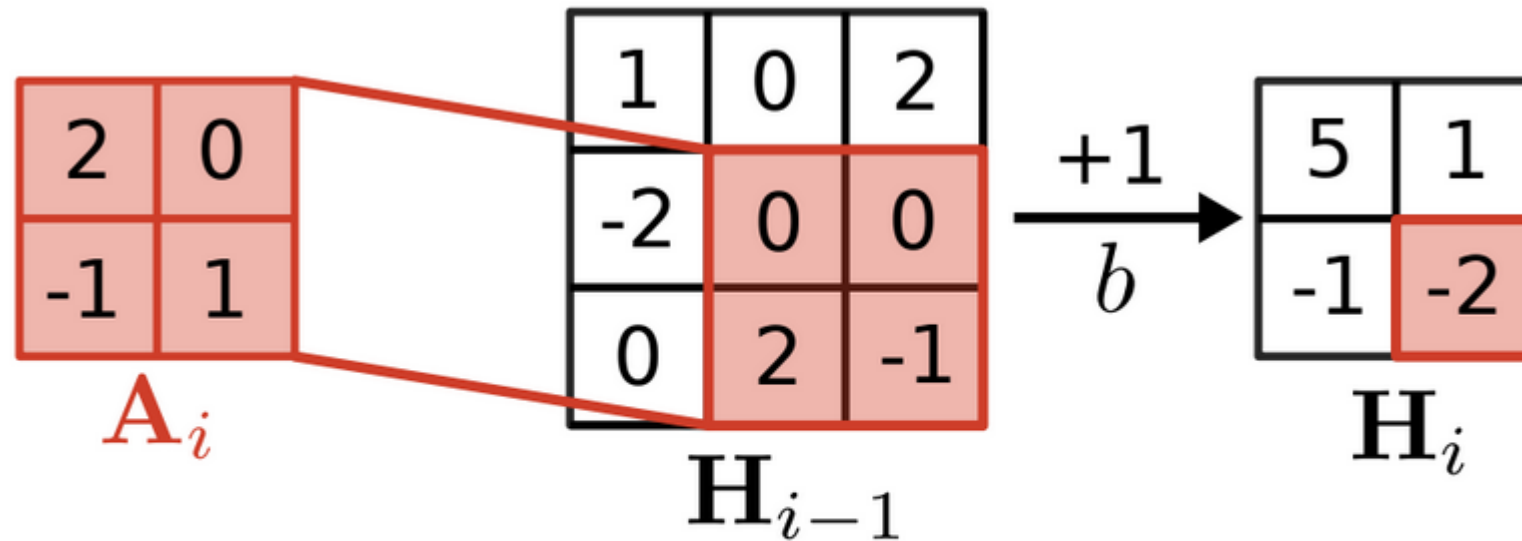


# Convolution Example



$$2 \cdot (-2) + 1 \cdot 2 + 1 = -1$$

# Convolution Example



$$(-1) \cdot 2 + 1 \cdot (-1) + 1 = -2$$

# Structure of this Lecture

## Network Architectures and Applications

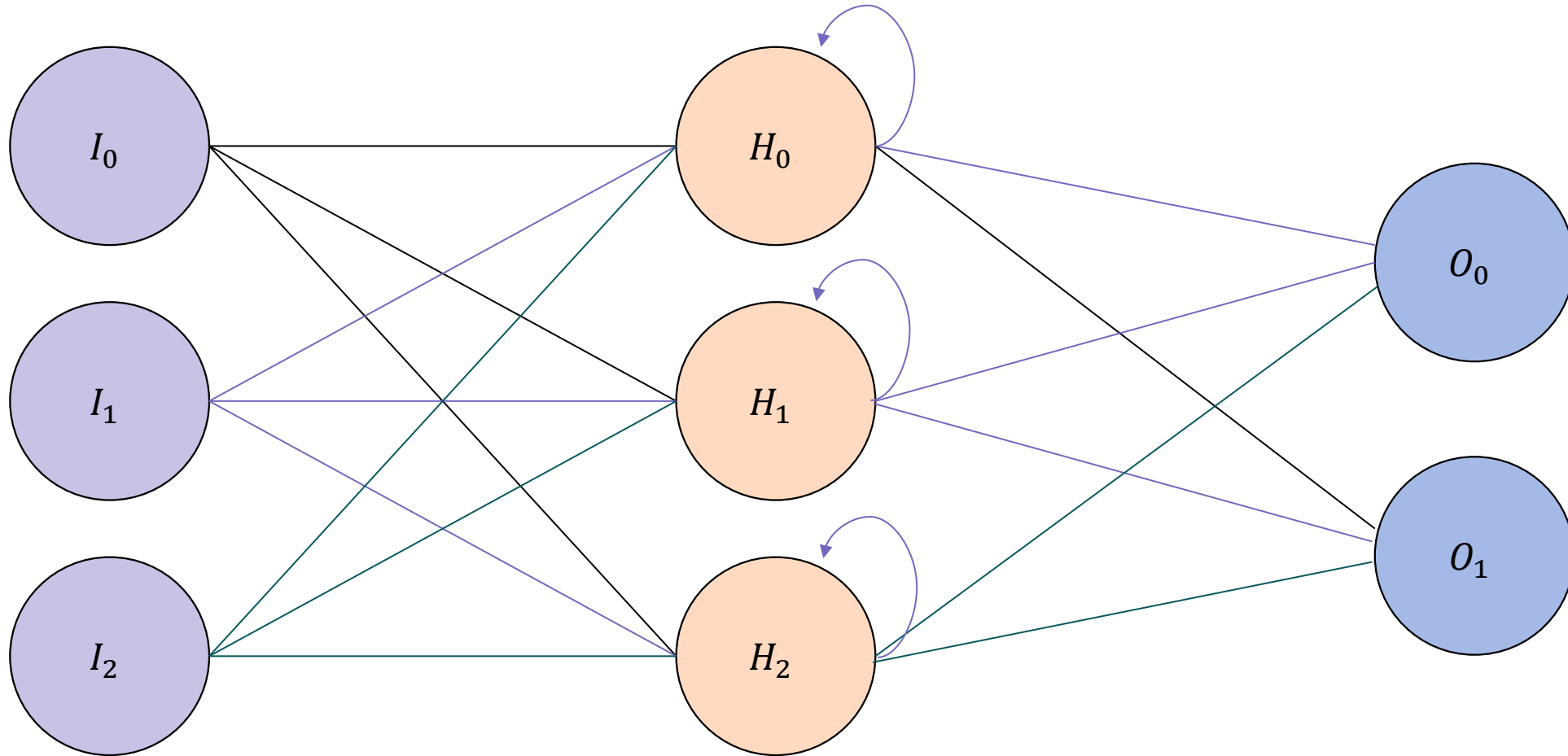
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# Sequences of Variable Length - Applications

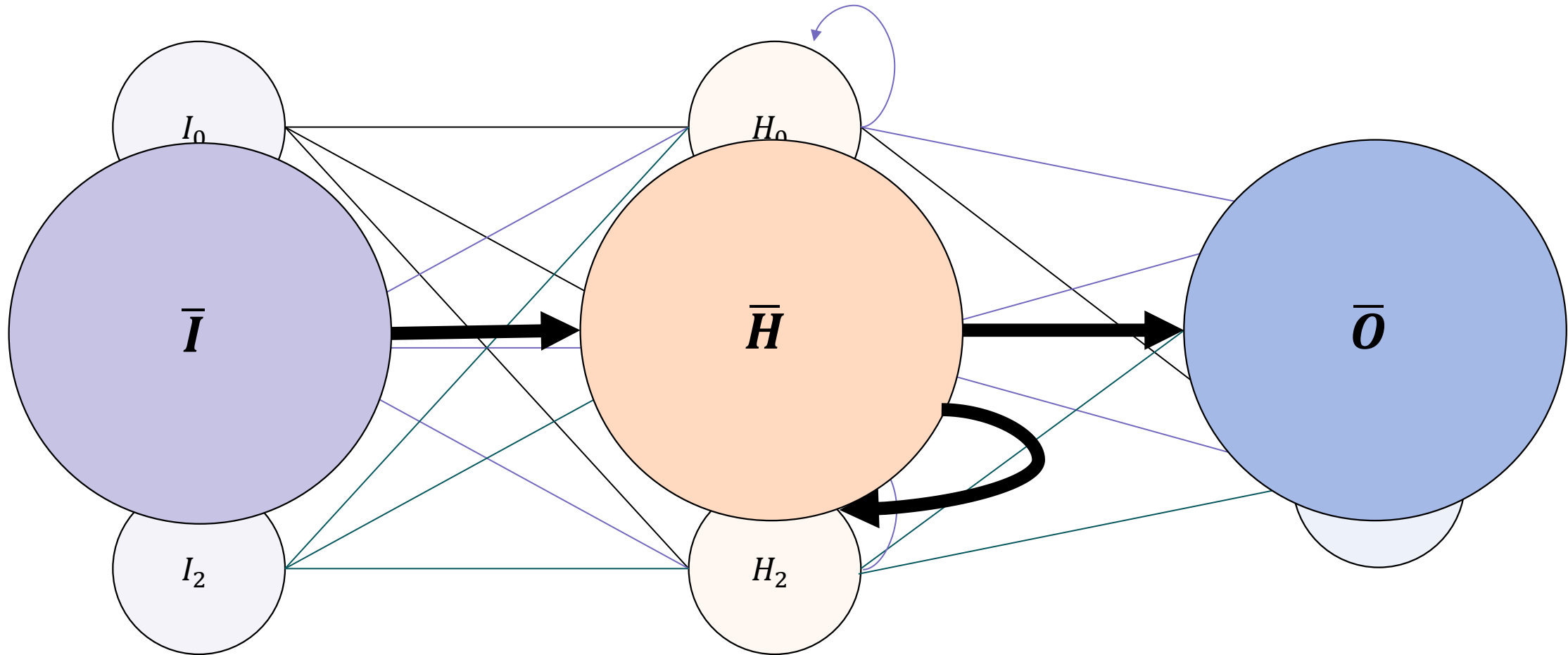
- Multiple object recognition
- Image segmentation
- Object tracking
- Image generation
- Image annotation
- Image captioning
- Machine translation
- Text generation

Each time step of process depends on „history“ (sequence of variable length)

# Recurrent Neural Networks

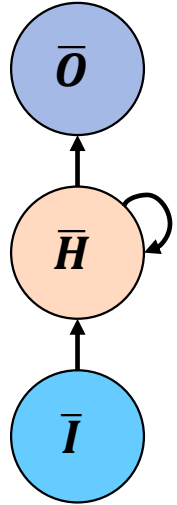


# Recurrent Neural Networks

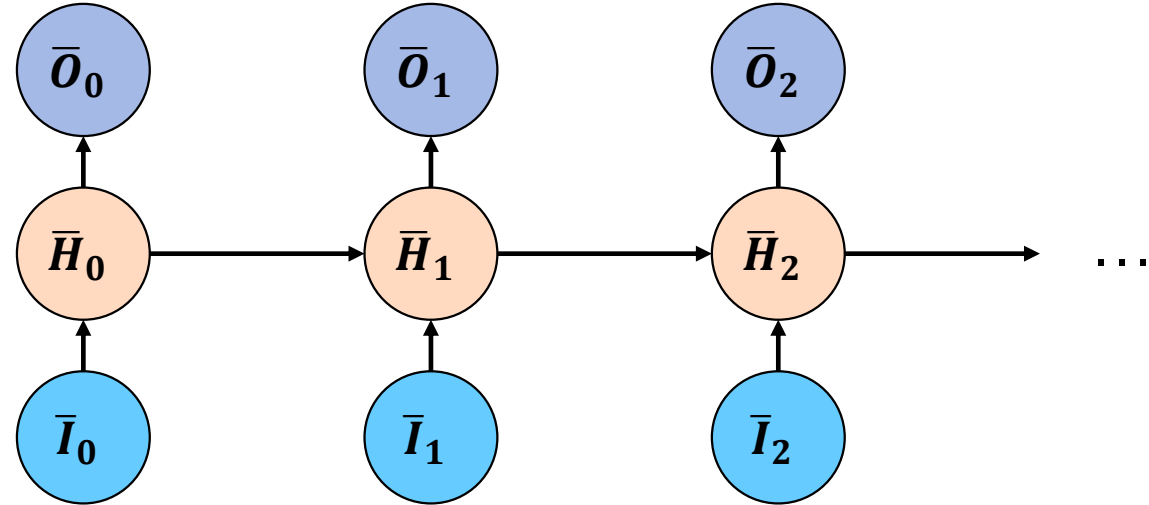


# Recurrent Neural Networks (RNN)

RNN with feedback connection



RNN unrolled over time (index = time  $t$ )



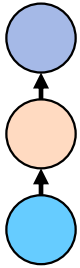
Update hidden state  $H$  based on **input and previous hidden state** using **same update rule** (same/shared weights and activation function) at each **time step**

Allows for processing sequences of variable length e.g. text data, not only fixed-sized vectors

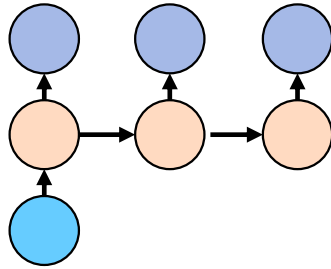
Infinite memory:  $H$  is function of all previous inputs (long-term dependencies)

# Recurrent Neural Networks – Mapping Types

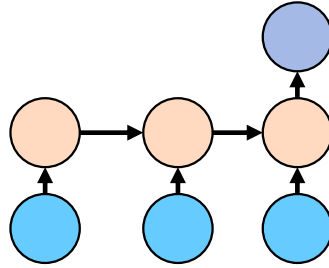
One to One



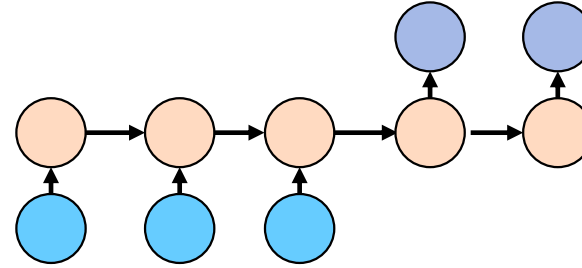
One to Many



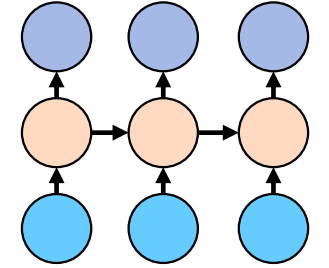
Many to One



Many to Many



Many to Many



RNNs allow for processing variable length inputs and outputs:

One to Many: Image captioning (image to sentence, music generation)

Many to One: Action recognition (video to action)

Many to Many: Machine translation (sentence to sentence)

Many to Many: Object tracking (video to object location per frame)

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# Neural Probabilistic Language Models – Word Representation

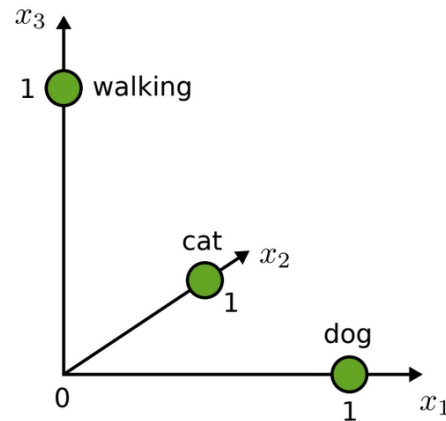
Dictionary  $D = \{a, cat, dog, have, walking\}$

Word-embedding

One-hot encoding

$a, cat, dog, have, walking$

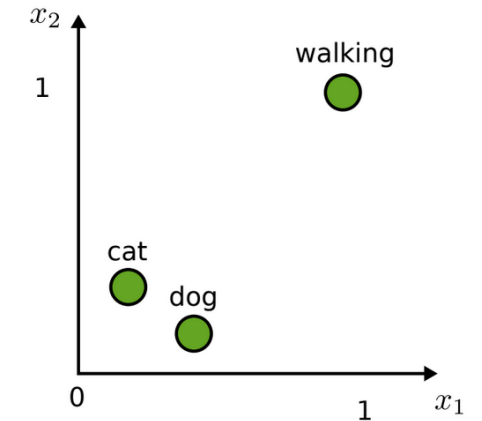
|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 |



- sparse
- high-dimensional
- hard-coded

$a \quad cat \quad dog \quad have \quad walking$

|      |      |      |      |      |
|------|------|------|------|------|
| 0.8  | 0.03 | 0.15 | 0.05 | 0.4  |
| 0.05 | 0.8  | 0.2  | 0.5  | 0.3  |
| 0.06 | 0.15 | 0.6  | 0.4  | 0.25 |
| 0.09 | 0.02 | 0.05 | 0.15 | 0.05 |

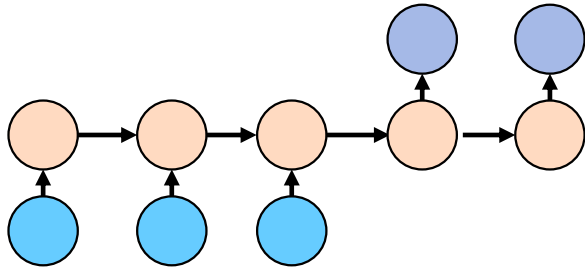


- dense
- lower-dimensional
- learned from data

Bengio et al.: A Neural Probabilistic Language Model, Journal of Machine Learning Research, 2003

# Text Data – Processing Sequences of Words

## RNN



## Self-Attention [1]

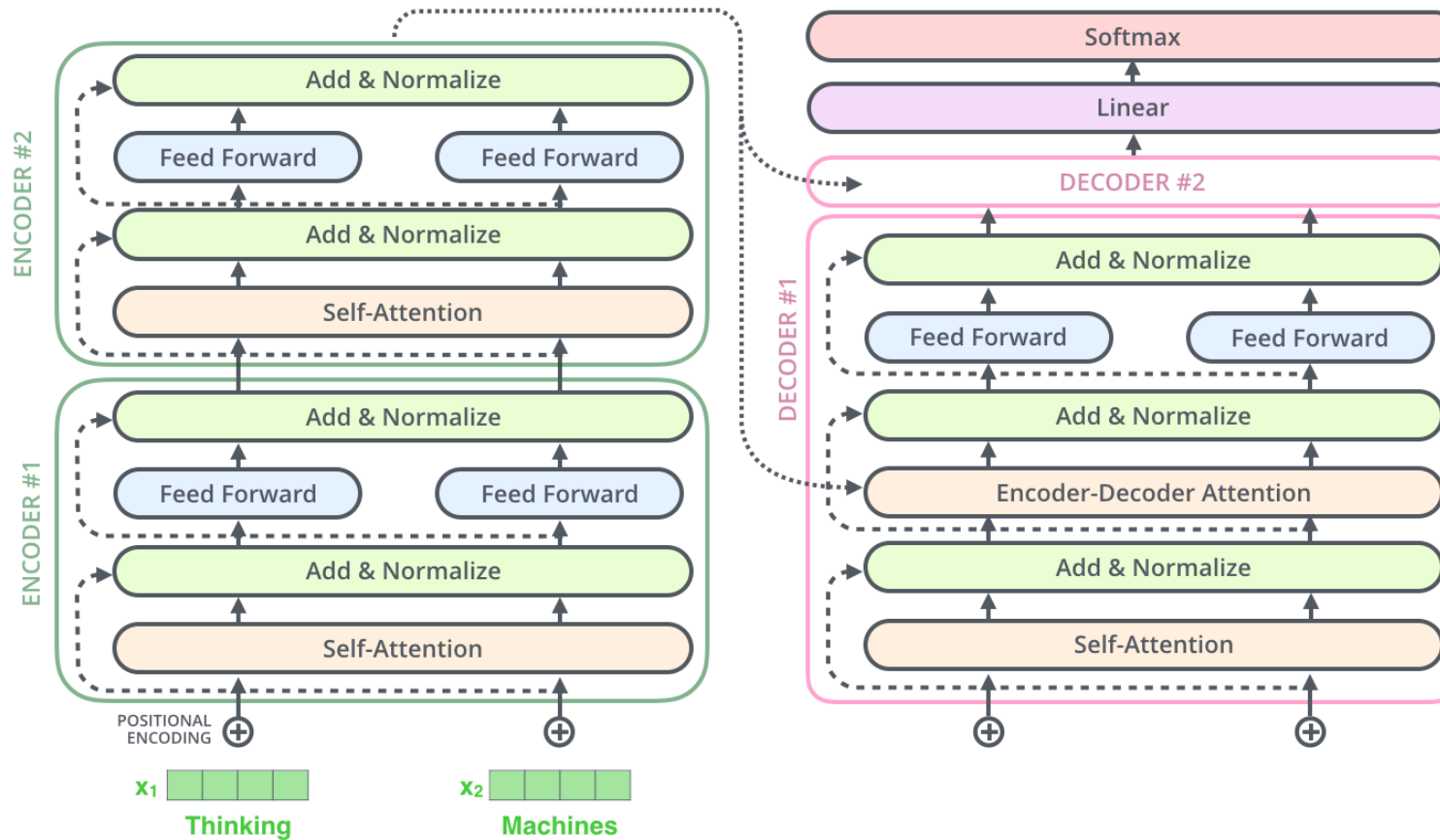
relates all tokens of a sequence with each other  
computes a representation of the sequence

can more easily **capture long-distance dependencies** compared to a RNN

calculation can be parallelized

[1] Vaswani et al. : Attention is all you need, NIPS 2017

# Deep Neural Network – Transformer Architecture

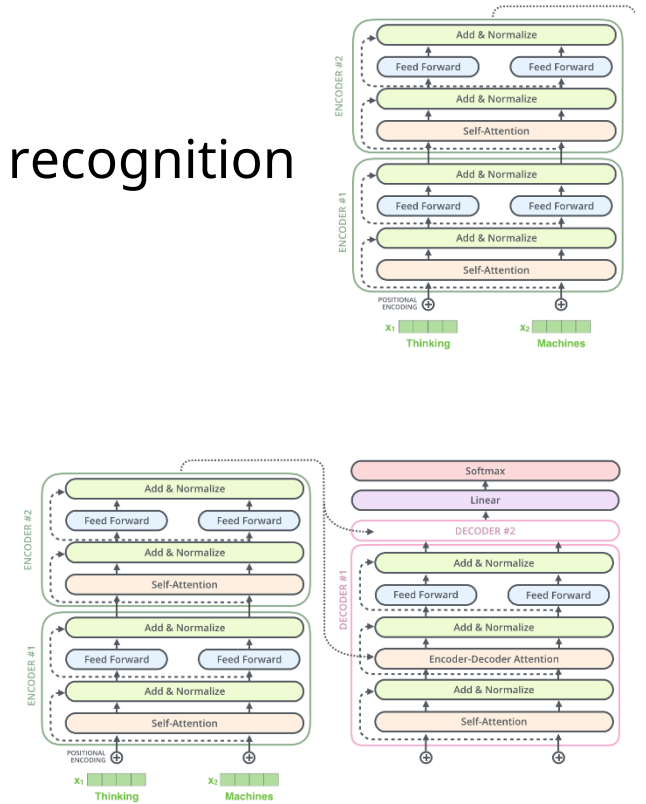
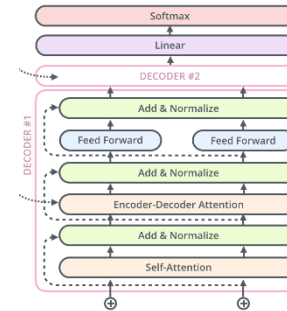


<https://jalammar.github.io/illustrated-transformer/>

Tutorial: <http://nlp.seas.harvard.edu/annotated-transformer/>

# Applications and Models

- Encoder Models  
e.g., Bert model [1], used for e.g. text classification, named entity recognition
- Decoder Models  
e.g., GPT-X [2], used for e.g. text generation
- Encoder-Decoder Models  
e.g., T5 [3], used for e.g. translation tasks



Explanation and implementation: <https://huggingface.co/learn/nlp-course/>

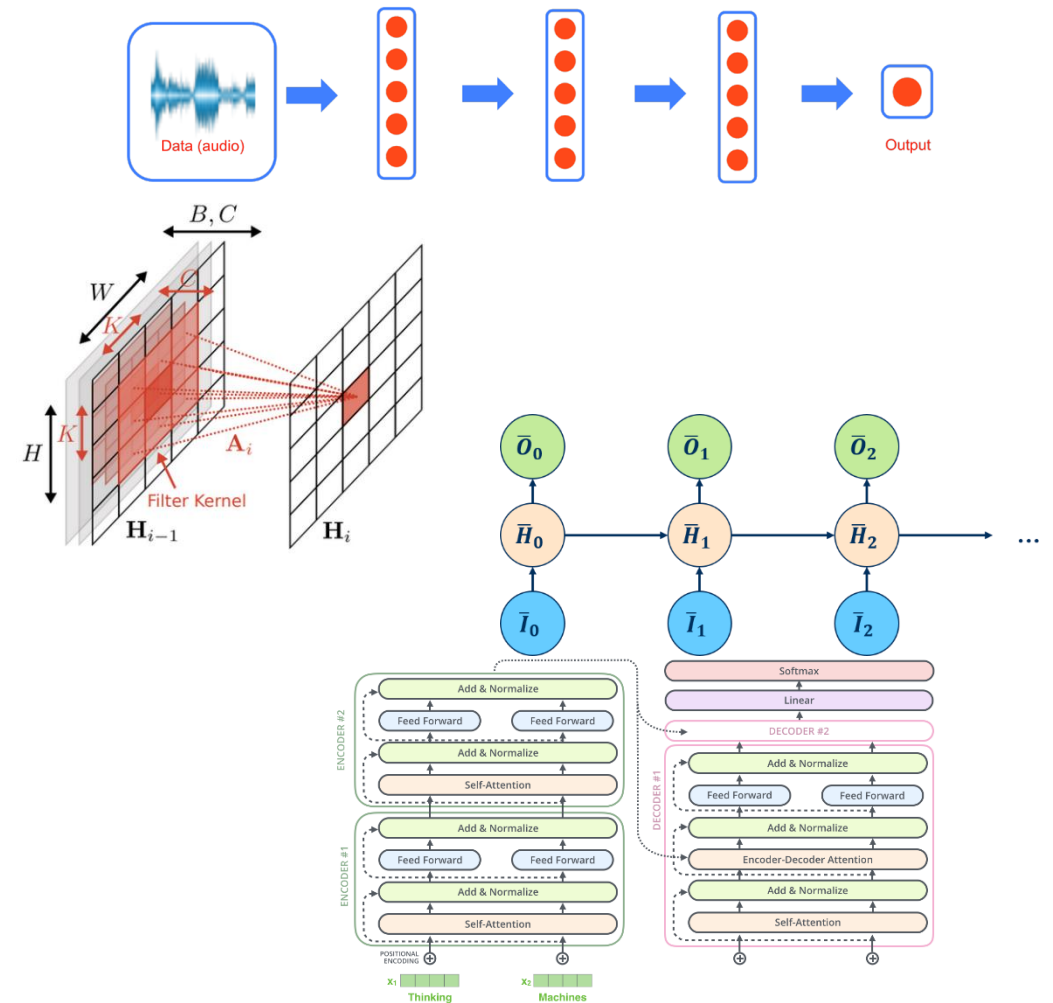
[1] Devlin et al.: BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, NAACL,HLT 2019

[2] Radford and Sutskever: Improving Language Understanding by Generative Pre-Training, arxiv, 2018

[3] Raffel et al.: Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer, Journal of Machine Learning Research, 2020

# Deep Learning Architectures and Applications and - Takeaway

- Deep Neural Networks allow to learn processing steps, which has been hand-crafted before
- Convolutional Neural Networks allow to share weight and thus reduce number of parameters
- Applied for image processing
- Recurrent Neural Networks allow to process sequences and take history into account
- Transformer based architectures are SotA for NLP
  - „Attention is all you need“
  - Encoder, Decoder and Encoder-Decoder models depending on NLP task



# Summary - Machine Learning

- Introduction to Machine Learning
- Similarity and Distances
- Clustering
- Supervised Learning – Training and Learning
- Classification
  - Decision tree
  - Bayes Classification Methods
- Model Evaluation and Selection
- Dimension Reduction
- Linear Regression
- Logistic Regression
- Deep learning



# Further References

## Convolutional Neural Networks and Recurrent Neural Networks

- <https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/>

**Lecture 7** Convolutional Neural Networks and **Lecture 08** Sequence Models

## Goodfellow, I. et al., *Deep Learning*, MIT Press, 2015

- <https://github.com/janishar/mit-deep-learning-book-pdf>

## Deep Learning Lecture Prof. Andreas Geiger, University Tübingen

- <https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/>

## Dive into Deep learning

- <http://d2l.ai/index.html>

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